WORK, POWER AND ENERGY

TIME: 1HR **MARKS:40**

SECTION A

[*All* *questions are compulsory*]

Question no.1: **[5]**

[i] In an oscillating pendulum the total energy at any instant of its motion is.....

[a] conserved

[b] sum of kinetic and gravitational potential energy

[c] total mechanical energy

[d] all of the above.

[ii] The energy conversion taking place when wax candle burns are.....

[a] chemical to light

[b] heat to light and chemical

[c] chemical to light and heat

[d] heat to light.

[iii] If a body of mass 10kg is lifted from the ground level to a height of 10m then the ratio of its inertia is .....

[a] 1:2

[b] 2:1

[c] 1:1

[d] 5:1

[iv] A body possesses kinetic energy X J. If the mass of body increases 49 times determine the final velocity if kinetic energy remains constant.

[a] $1/5^{th}$ the original velocity

[b] $1/14^{th}$ the original velocity

[c] $1/7^{th}$ the original velocity

[d] $1/10^{th}$ the original velocity.

[v] Power rating for a centrifugal pump is.....

[a] work output / work input

[b] power output / power input

[c] work input / work output

[d] both a and b

Question 2: **[5]**

[i] In a kitchen mixie energy conversion taking place is electrical energy into ..... [kinetic energy/ mechanical energy/ mechanical kinetic rotational energy]

[ii] Amount of work done when force is applied is ..... [change in kinetic energy/ change is gravitational potential energy / both]

[iii] The heart of a rabbit beats 100 times in a minute when a fox chases it during which the work performed is 1.5J. The power of heart is ..... [25W / 0.25W / 2.5W]

[iv] A weight lifter and coolie raises a load of 500N to a height of 6m in 10 min and 20min respectively. The ratio of power spend by weight lifter to the power spent by coolie is ..... [5:2/ 2:5 / 2:1]

[v] A body of mass 0.5kg travel rectilinearly with velocity equal to $ax^{3/2}$.If acceleration of the body is $5ms^{-2}$ the work done by net force during its displacement from x=0m to x=2m is... [50J / 100J/200J]

Question 3:

a] State factors on which the amount of work done depends? **[2]**

Ans: The amount of work done depends on:

[i] Force applied by the body.... Greater the force more is the work done by the body.

[ii] Displacement in the body: Greater the net displacement in the body greater is the work done.

b] Derive the relationship between 1kilowatt-hour and 1 joule. **[2]**

Ans: Using the relationship,

1 kilowatt-hour	**=**	**1kW x 1h**
	=	**1000 W x 3600s**
	=	**1000 Js^{-1} x 3600s**
	=	**3.6 x 10^6 J**

c] An electric motor of power 100W is used to drive the stirrer in a water bath. If 50% of the energy supplied to the motor is spent in stirring the water. Calculate the work done on water in 1 minute**.** **[2]**

Ans: Power of motor is 100 W	**=**	**100 Js^{-1}**
Time of usage of motor	**=**	**1 minute**
	=	**60 second**
Energy supplied to the motor	**=**	**Power x time**
	=	**100 x 60**
	=	**6000J**
Given only 50% energy is utilized	**=**	**50% of 6000**
	=	**3000J**

Hence the amount of work done on water is 3000J

d] A force of 10N displaces a body by 2 m at an angle of 60° to its own direction. Find the amount of work done? **[2]**

Ans: Amount of work done = $F\cos\theta \times S$

= $10 \cos 60^o \times 2$

= $10 \times ½ \times 2$

= 10J

e] State the work energy theorem. **[2]**

Ans: Work energy theorem states that when a force is applied on a body the amount of work done is equal to change in kinetic energy or is also equal to change in gravitational potential energy. Mathematically,

Work done = $½ mv^2 – ½ mu^2$

= $mgh_2 – mgh_1$

SECTION B

[*Attempt **any two** questions out of three]*

Question no.3:

a] [i] For the same kinetic energy of the body, what should be the change in velocity if its mass is increased to four times its original mass? **[2]**

Ans:

	Case 1	Case 2
Mass	M kg	4M kg
Velocity	V_1 ms^{-1}	V_2 ms^{-1}
Kinetic energy	KE	KE

Given Kinetic energy in both cases is constant.

Applying the above condition we have,

$$\frac{1}{2} mv_1^2 = \frac{1}{2} mv_2^2$$

$$\frac{1}{2} mv_1^2 = \frac{1}{2} [4m] v_2^2$$

$$V_1^2/V_2^2 = 4m/m$$

Ratio of initial velocity to final velocity is 2:1which means final velocity is halved the initial velocity.

[ii] Obtain the correlation between power force and velocity **[2]**

Ans: By the definition of power,

$$\text{Power} = \text{Work done / time}$$

$$= \text{[Force x displacement] / time}$$

$$= \text{Force x [displacement / time]}$$

$$\text{Power} = \text{Force x velocity.}$$

b] A body of mass 5 kg is taken from a height 5m to 10m. Find the increase in gravitational potential energy. [Take g =10ms^{-2}] **[3]**

Ans:

Increase in gravitational potential energy = $mgh_2 - mgh_1$

$$= mg [h_2 - h_1]$$

$$= 5 \times 10 \times [10 - 5]$$

$$= 250 \text{ J}$$

c] A body falls freely under gravity from rest. Name the kind of energy it will possess,

[i] at a point from where it is just about to fall

[ii] while falling.

[iii] on striking the ground. **[3]**

Ans: The energy possessed at:

[i] point where it just about to fall is Gravitational potential energy.

[ii] While falling it possess both Kinetic energy and gravitational potential energy.

[iii] On striking the ground the energy possessed by the body is kinetic energy.

Question no.4:

a] A truck driver loads some oil drums into a truck by lifting them directly. Each drum has a mass of 80kg and the platform of the truck is 0.8m above the ground.

[i] How much force is needed to lift a drum into the truck?

[ii] How much energy is used up in lifting a drum?

[iii] After the truck is loaded the driver drives off. List the major energy changes that take place in moving the truck?

[iv] The driver stops the truck suddenly at the factory gate. What happens to the kinetic energy of the truck? **[4]**

Ans:

[i] Force needed to lift the drum into the truck is equal to weight of the drum = mass x acceleration due to gravity

= [80 x 10] ..... assuming g = 10ms^{-2}

= 800N

[ii] Energy used in lifting a drum is the Gravitational potential energy = mass x acceleration due to gravity x height.

= [80 x 10 x 0.8]

= 640J

[iii] Major energy changes taking place when the truck starts are chemical energy to heat energy to mechanical kinetic energy.

[iv]When the truck stops suddenly at the factory gates the mechanical kinetic energy gets converted into mechanical potential energy and heat energy due to friction between tyres and road.

b] In what way does the temperature of water at the bottom of a waterfall differ from the temperature of the top? Explain briefly**[3]**

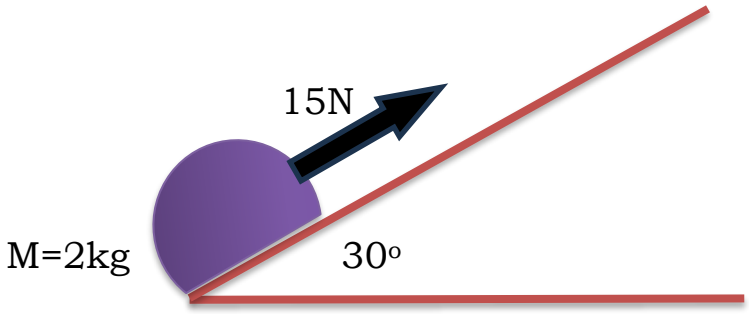
Ans: The temperature of water at the bottom of waterfall would slightly be greater than temperature of water at the top of the dam as when the water falls the gravitational potential energy possessed by the water at the top gets converted into mechanical kinetic rotational energy of the turbine blades. Along with the rotation some amount of heat is also generated due to frictional resistance of the water and turbine blades thus increasing the temperature of water at the bottom of waterfall.

c] When a body moves in a circular path what is the work done by it when it covers quarter circle, half circle and three fourth circle.**[3]**

Ans: If the frame of reference considered is angle between force and displacement then in all three cases the work done would be zero as displacement along the periphery is always tangential to the centripetal force acting radially inwards towards the centre of the circular path. Hence cosθ would be zero. So work done would be zero.

Question no.5:

a] A force of 15 N is required to pull up a mass 2kg through 5m along an inclined plane making an angle of 30° with the horizontal as shown in the figure.



Calculate:

[i] The work done by the force in pulling the body along the plane.

[ii] The force due to gravity on the body.

[iii] The work done against the force of gravity [g= 9.8 ms^{-2}]

[iv] Power spend in pulling the body along the plane in 15 sec. **[4]**

Ans:

[i] The work done in pulling the body along the plane is obtained by the relation Force x displacement.

= [15 x 5]

= 75J

[ii] The force due to gravity is the weight of the body.

= mass x acceleration due to gravity.

= [2 x 9.8]

= 19.6 N

[iii] Work done against force of gravity is the gravitational potential energy.

= mass x acceleration due to gravity x height

= [2 x 9.8 x 5 x sin 30°]

= 49J

[iv] Power spent in pulling the body along the plane

= work done / time

= [75 / 15]

= 5 watt.

b] Obtain the correlation between Kinetic energy and momentum.**[3]**

Ans: Using the relation,

Kinetic energy = ½ x mass x [velocity]^2

and linear momentum = mass x linear velocity

Kinetic energy = ½ [mass x velocity] x velocity

Now, K.E. = 1/2m x {[mass]^2 x [velocity]^2}

2m x K.E. = [linear momentum]^2

Linear momentum = $\sqrt{}$2m x K.E.

c] A stone of mass 64 g is thrown vertically upwards from the ground with an initial speed of 20ms^{-1}. Calculate the potential energy at the maximum height attained by the stone. [Take g = 10ms^{-2}]. State what would be the total energy of the body at its half-way point? **[3]**

Ans: By law of conservation of energy,

Kinetic energy at the bottom = Potential energy at the top

Hence,

Potential energy at the top = ½ x mass x [velocity]^2

= ½ x [64 / 1000] x [20]^2

= 128J

Hence the Potential energy attained by the stone at the maximum height is 128J.

The total energy of the stone at any instant would be equal to 128J.

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